Kushagra Karar

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Research interests: Constrained and structured decoding, reliability of LLM/VLM outputs, ML Systems, LLMs for Software Engineering

Education

Indian Institute of Technology (IIT) Delhi — New Delhi
M.S. (Research) in Computer Science & Engineering – Advisor: Prof. Abhilash Jindal — *2024 – Present*

Indian Institute of Information Technology (IIIT) Una — Una, Himachal Pradesh
B.Tech in Computer Science & Engineering; CGPA: 8.66 — *2020 – 2024*

Publication

K. Singh, **K. Karar**, A. Jindal, G. Yang. *Speculate: Generating REST API Specifications Using LLMs.* ACM International Conference on the Foundations of Software Engineering (**FSE 2026**). doi.org/10.1145/3797118 [**Blog**] **ACM Artifacts Available + Evaluated–Reusable.**

Research

Cross-field semantic constraints via SMT solving in VLM beam search — Dec 2025 – Present
IIT Delhi (with Prof. Abhilash Jindal) — *Lead researcher*

- Designed a generic **cross-field semantic constraint framework** that embeds **SMT solving** directly inside **token-level constrained beam search**, enforcing inter-field dependencies *during* decoding
- The framework generalizes across a variety of document datasets such as **MIDV-2020** (Passport and ID documents) and **CORD**(Restaurant Receipts), covering **4 distinct constraint types** , yielding F1 gains of up to **+16** points.
- Achieved a speedup of **33%** through **GPU-CPU overlaps** via **batch parallelism** by creating individual **stateful processes.**

Speculate: Generating REST API Specifications Using LLMs — Aug 2024 – Sep 2025
IIT Delhi + University of Queensland (with Profs. A. Jindal & G. Yang) — *Co-author*

- Speculate is a modular system that **combines lightweight static analysis with LLMs** to automatically generate REST API specifications from source code.
- Speculate generalizes across multiple languages and frameworks by successfully evaluating across **19 real-world repositories** spanning Python and Java (**Django, Jersey, and Spring Boot**).
- Speculate outperforms state-of-the-art static analysis tools on both **precision and recall** across five evaluation metrics across the 19 repositories.

Talks & Posters

Talk: *Speculate: Generating REST API Specifications Using LLMs* – ACM FSE 2026, Montréal, Canada [Link] — Jul 2026

Poster: Speculate – IIT Delhi Open House 2026 [Link] — Oct 2026

Poster: Speculate – Google Day @ IIT Delhi — Oct 2026

Skills

ML / LLMs: Python, PyTorch, Transformers, vLLM, constrained decoding (xGrammar), KV caching, Paged Attention, Prefix Caching, Speculative Decoding
Systems & Software: C++, Data Structures, Object Oriented Programming, Operating Systems, Database Management Systems, Theory of Computation, REST APIs, Linux, Docker, Git, Z3 SMT solvers, Static analysis

Other

Mobile App Developer Intern, PhyFarm [Certificate] — Dec 2022 – Apr 2023

SeatScore: YOLOv11 transfer-learning project for fatigue detection [Code]

References

Prof. Abhilash Jindal, IIT Delhi

Prof. Guowei Yang, University of Queensland